

Redis 4.0.0 introduces sophisticated replication

Redis, the popular in-memory open source database, has reached v4.0.0. The new version has eliminated some of the known limitations of its previous version and includes new upgrades.

One of the most noticeable changes in Redis 4.0.0 is the addition of a new replication engine called PSYNC2. This enables the slaves promoted to masters to accept the old slaves without a full resynchronisation. In the same way, slaves can be stopped, upgraded and restarted with a partial resynchronisation.

The Redis team has also added active memory defragmentation to make the experience seamless. You can scan the memory key space allocations and rearrange the tablet structure to reduce memory fragmentation. Additionally, the new Redis version can work with the default defragmentation system on Linux.

"I am very happy about 4.0.0, because this release finally overcomes several very important limits of Redis," Salvatore Sanfilippo, the Redis creator, wrote in a Google Groups post.

In addition to the new replication and memory control functions, the new Redis version has modules to extend database functions. Redis users are also enthusiastic about the eviction algorithm that is used to clean the database cache.

To cater to the growing demand for Raspberry Pi in developing new IoT projects, Sanfilippo has added official Raspberry Pi support. "Every new Redis release is tested on the Pi," he wrote, adding that the Redis code was audited and fixed for unaligned accesses, while stack traces are supported on ARM hardware.

Various changes to Redis Cluster are planned for the next release. Meanwhile, you can use Redis 4.0.0 on your system and experience the benefits of all the subtle tweaks.

SparkyLinux 5.0 brings out Debian 'Buster'

After bringing out the latest Debian support last month, SparkyLinux has been updated to version 5.0 with the upcoming testing branch 'Buster' on board. The new update, codenamed Nibiru, is available with LXQt, MATE and Xfce desktop environments. Unlike the previous Debian version, SparkyLinux 5.0 upgrades the system using the Debian testing repositories that enable Debian 10 'Buster'. The distribution is yet to be released through an official channel. However, security testing platform Parrot Security OS brought out Debian 10 support a few weeks ago.

Apart from the upcoming Debian version, SparkyLinux 5.0 has Linux 4.11.6 as the default kernel and comes with a new Sparky5 theme that includes fresh wallpapers. The operating system also replaces the original Thunderbird client with Icedove and has Calamares 3.1.1 as the default system installer. Similarly, there is a new tool to check and generate push notifications on the availability of new updates. Developers have been provided with a total of five SparkyLinux variants — three of them are available as live images, while the other two come in the MinimalGUI and MinimalCLI versions. Also, the SparkyLinux team has supported the ongoing HTTPS transition. Therefore, it enables the open source platform to use the HTTPS protocol for all Sparky services.

You can download SparkyLinux 5.0 from its official website. If you already have Sparky 4.6, you can upgrade the old version by typing the 'sudo apt-get update' and then the 'sudo apt-get dist-upgrade' commands in the terminal.

OpenText takes on IBM's Watson by launching AI platform Magellan

OpenText, the Canada-based enterprise information management company, has entered the emerging world of deep learning by building its own native artificial intelligence (AI) platform called Magellan. The new development is an extension to OpenText's specialty — designed to compete against IBM's Watson. Waterloo-headquartered OpenText is employing the open source approach for Magellan by integrating Apache Spark and MLlib. This approach is aimed to help customers leverage open source algorithms for building their own models. The biggest value proposition of Magellan is the lower price point. Further, OpenText is launching Magellan as an appliance, cloud service and for the local server to meet a variety of requirements.

"Magellan enables machine-assisted decision-making, helping you automate and optimise your business operations. You can gain value and insight from your data stores with Magellan, and leverage the benefits of AI-powered analytics for faster decision-making and task automation," the company blog post reads.

Magellan blends the text mining and auto classification capabilities from Nstein, whereas the analytical technologies are mainly driven by Actuate. OpenText acquired both Nstein and Actuate last year. The prime ability of OpenText to take unstructured and semi-structured data for AI using an open source technology stack helps it take on Watson. Magellan is targeted at mid-sized and large companies that do not have an internal team of data scientists. The ability to deploy text mining and analysis capabilities to work on unstructured data distinguishes the open source-powered offering in the market. OpenText assures customers that they will save on training and integration by using its solution.

Magellan is already touted to be well-optimised for analysing proposal requests. Users can leverage it to take unstructured data in RFPs and analyse the exact requirements of customers.

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